NHH

HOME EXAMINATION

BAN404

Spring, 2022

Date: 19.05.2022

Time: 8.00-16.00

Number of hours: 8

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH.

SUPPLEMENTARY REGULATIONS TO HOME EXAMINATIONS

You can find supplementary regulations under the headline "Regulations"

https://www.nhh.no/en/for-students/regulations/

Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: 3

Instructions

Your solution should be submitted as two files.

1. A pdf-file where both your R-code and the answers to the tasks are given.
2. Either a well documented R-script (.R) or an R markdown file (.Rmd) which can reproduce the results in the pdf-file.

The datasets used here are well analyzed before. Make sure to refer to other's work if you use it to solve the tasks.

A general recommendation: Be pragmatic if you run into difficulties. You need to find models that can produce predictions. Don´t let the perfect prediction stand in the way for a good one!

The number in brackets after each subtask shows the maximum score for that subtask.

Throughout the exam you will work with training and test data and you will have to make appropriate choices on when to use one or the other. In cases when you think there are more than one valid choice on working with the training, test or the full dataset, explain your choice!

Task 1

In this task you will look at the `OJ` data in the `ISLR` package. The dataset contains 1070 purchases of two brands of juice. In the task you should analyze and predict the variable `Purchase`, a variable showing which of the two brands, Citrus Hill (CH) or Minute Maid Orange (MM), was purchased. See the help-function to the dataset for more information of the dataset. Except in Task f), you should assume that the variable `LoyalCH`, a variable indicating loyalty to the Citrus Hill brand, is unavailable at the time of prediction.

a) Remove the variable `LoyalCH`. If necessary, recode categorical variables as factors and remove variables that cannot be used in the analysis. Base your reasoning to do this on the help function and by looking at the data. Split the data in a 50/50 split to create a training and a test dataset. Use the seed 8655, when you split the data, `set.seed(8655)`. (4)

b) Use descriptive statistics to detect promising predictors for `Purchase`. Present the most interesting results as tables and graphs and comment on them. (4)

c) Fit a logistic regression of `Purchase` on all the variables you found promising in b), evaluate the predictions using test accuracy and a confusion matrix. If you, in the process, conclude that you would modify your model, you should do so and explain why. Use the threshold 0.5 when you go from estimated probability to a prediction. (8)

d) Fit a classification tree, using all variables (except the one(s) you removed in a), and plot it. Give an example of how to interpret a branch of the tree (How to predict a new observation with certain values on the explanatory variables). Also, use the same evaluation measures as in c) and evaluate the predictions. Use the threshold 0.5 when you go from estimated probability to a prediction. (5)

e) In the c) and d) you have used the threshold 0.5. Use cross-validation to find the threshold that maximizes the accuracy, i.e. the fraction of correctly predicted observations. Evaluate predictions from the tree using the best threshold. Are the predictions improved? Use the seed 4598, when you split the data, `set.seed(4598)`. (15)

f) In a) you removed the variable `LoyalCH`. Imagine that this variable is actually available at the time of a prediction. By augmenting your analysis with this variable both for the logistic regression and the classification tree, conclude if this variable contribute to predicting `Purchase`. (8)

Task 2

In this task you will work with the dataset `Computers` in the `Ecdat` package. You should predict the variable `price` using properties of the computers and other explanatory variables. In the process of this we would also like to learn something about the properties of low- and high-priced computers.

a) If necessary, recode categorical variables as factors and remove variables that cannot be used in the analysis. Base your reasoning to do this on the help function and by looking at the data. Split the data in a 50/50 split to create a training and a test dataset. Use the seed 4598, when you split the data, `set.seed(4598)` (4)

b) Use descriptive statistics to find promising predictors. Comment on your observations. (4)

c) Run a linear regression on all explanatory variables. Interpret some of the coefficients. Evaluate the prediction performance on an appropriate measure. (5)

d) Investigate if the the predictions of `price` would be better if we use a linear regression of `log(price)` on the other variables. (7)

e) In d) you fitted a model where `price` is a particular nonlinear function of the other variables. You should now investigate another non-linear models. First, fit a GAM-model, plot the result and evaluate the predictions. Is there a reason to not allow for some variables to have a non-linear relationship with `price`? (12)

f) Give a brief explanation of how backfitting works and in what situations it is possible to fit a generalized additive model with ordinary least square regression. (7)

g) Use bagged trees to predict `price`. Evaluate the predictions. Compute variable importance measures and comment on the results. Explain the logic behind the measure for variable importance that you are using. (Hint: If the computation takes a long time you can reduce the number of trees fitted using the argument `ntree`.) (10)

h) Give a brief explanation of bagging. (7)